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PiGIE: Proximal policy optimization guided diffusion for fine-grained image editing

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ABSTRACT

Instruction-based image editing is a challenging task since it requires to manipulate the visual content of images according to complex human language instructions. When editing an image with tiny objects and complex positional relationships, existing image editing methods cannot locate the accurate region to execute the editing. To address this issue, we introduce *Proximal Policy Optimization Guided Image Editing* (PiGIE), a diffusion model that can accurately edit tiny objects in images with complex scenes. The PiGIE can incorporate proper noise masks to edit images based on the guidance of the target object's attention maps. Different from the traditional image editing approaches based on supervised learning, PiGIE leverages reinforcement learning with Proximal Policy Optimization (PPO) to fine-tune the diffusion model by using the cosine similarity between UNet attention maps and human feedback as the reward signal. On multiple image editing benchmarks, PiGIE exhibits remarkable improvements in both image quality and generalization capability. In particular, PiGIE sets a new baseline for editing fine-grained images with multiple tiny objects, shedding light on future studies on text-guided image editing for tiny objects.

1. Introduction

Instruction-based image editing is a challenging task since it requires to manipulate the visual content of images according to complex human language instructions. With great controllability and the strong ability to generate realistic images, diffusion models (Ho et al., 2020; Huang et al., 2025; Sohl-Dickstein et al., 2015; Song et al., 2021; Zheng et al., 2024; Zuo et al., 2025) have been widely used in applications such as image editing (Huang et al., 2024; Kim et al., 2022; Preechakul et al., 2022; Wang et al., 2023).

Usually, an instruction-based diffusion model takes both an input image and a human instruction prompt as inputs and can generate edited images according to input instructions. However, the training instruction-based diffusion model requires high-quality instruction data on a large scale. To address such a limitation, InstructPix2Pix (Brooks et al., 2023) proposes a practical and effective method to generate large amounts of instruction-images paired data for instruction fine-tuning diffusion models for image editing. The instructions are generated via GPT-3 (Brown et al., 2020), the input target image is synthesized using Prompt-to-Prompt (Hertz et al., 2022) image editing method, and the diffusion model takes both a text embedding of a pre-trained CLIP text-

encoder (Radford et al., 2021) along with the input image. Though InstructPix2Pix has gained satisfactory success, the quality of this dataset is limited by GPT-3's ability to create instructions and modify captions, as well as Prompt-to-Prompt modifications to the generated images, which makes InstructPix2Pix challenging to edit tiny objects and small regions in images with complex editing scenarios.

To address the limitations of editing tiny objects, we introduce *Proximal Policy Optimization Guided Image Editing* (PiGIE) in this work. This state-of-the-art instruction-based image editing diffusion model can edit tiny objects with complex spatial relationships in images. PiGIE acquired such a fine-grained editing ability by incorporating reinforcement learning with proper reward functions that emphasize the similarity between the attention map of the edited image and the human-perceived editing regions. With the incorporation of RL in the instruction fine-tuning, PiGIE shows remarkable ability and accuracy in editing tiny objects in complex images. As demonstrated in Fig. 1, the advanced ability of PiGIE to understand and locate the accurate regions to edit distinguishes PiGIE from the existing approach when editing complex images.

To train the PiGIE, we adopt the open-source dataset from MagicBrush (Zhang et al., 2024a) as our basic pre-training dataset. To simulate manual annotation, we choose the free-form masked images

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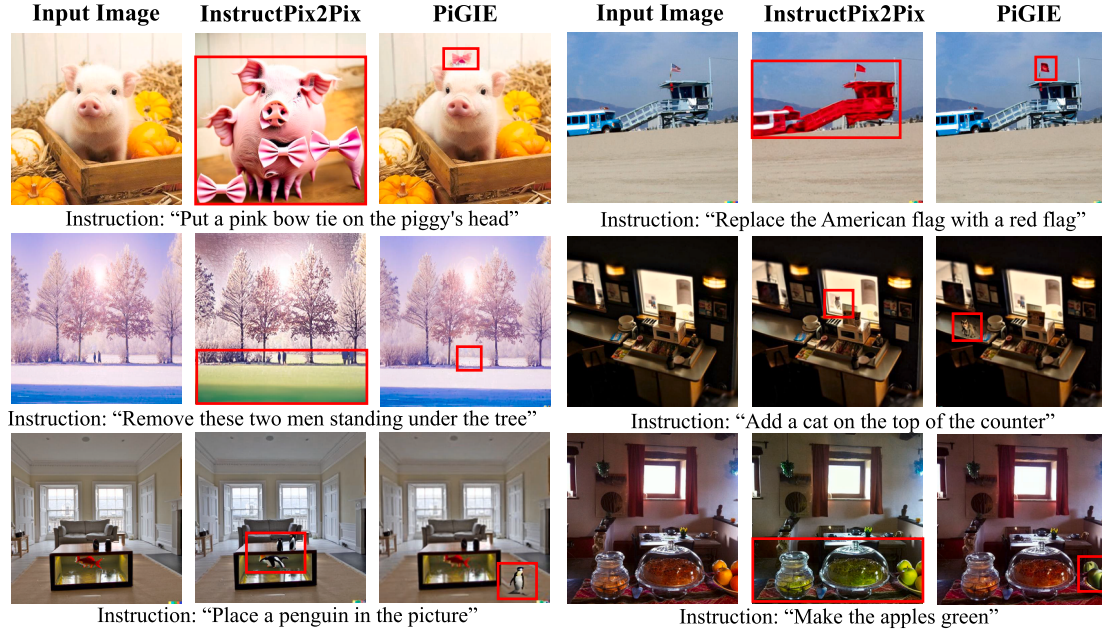


Fig. 1. We introduce PPO-Guided Image Editing (PiGIE) to optimize instruction-based image editing for various editing tasks. The bottom is the edit instruction, the left is the input image, the middle is the image edited by InstructPix2Pix (Brooks et al., 2023), and the right is the image after PiGIE editing. The red bounding box represents the editing area. Compared to InstructPix2Pix, our PiGIE accurately locates the editing area and achieves fine-grained localized editing.

from MagicBrush as human feedback. We elaborately compare different Instruction-guided editing methods in InstructPix2Pix (Brooks et al., 2023), MGIE (Fu et al., 2023) and several complex image datasets in PASCAL VOC (Everingham et al., 2010), COCO (Lin et al., 2014) and VG (Krishna et al., 2017). Through extensive experiments, we find that PiGIE exhibits remarkable advancements in both quality and generalization capabilities when tasked with fine-grained image editing operations, outperforming existing methods such as InstructPix2Pix across four evaluation metrics. We summarize the contributions of this paper in three points:

- We introduce a reinforcement learning framework for instruction-based fine-grained image editing, where the differences between attention maps are used to construct reward signals for policy optimization. This design enables the diffusion model to better align the editing process with the desired editing objectives, improving its adaptability to fine-grained editing tasks beyond the limitations of prompt-based methods.
- We propose a novel Attention Map Reward to measure the similarity between the attention maps of edited images and human-perceived editing regions. To approximate human feedback, we utilize the free-form masked images provided in the MagicBrush dataset to simulate the target editing regions.
- By incorporating Attention Map Reward and Clip Reward as components of the reward function, the proposed model can accurately localize editing regions in complex scenes, enabling fine-grained and instruction-consistent modifications. This design allows the model to support both stylistic editing and precise localized editing.

2. Related work

2.1. Instruction-based image editing

Text-guided image editing can significantly improve the flexibility and accessibility of the visual content of an image according to natural language commands. Previous work has edited images by using the GAN framework (Goodfellow et al., 2020; Reed et al., 2016), but it only edits

part of the region and is prone to produce unrealistic results (Fu et al., 2022; Li et al., 2020). With the emergence of diffusion models (Ho et al., 2020), significant progress has been made in controllable image generation and editing. Diffusion-based editing approaches include training-based methods (Yildirim et al., 2023; Zhang et al., 2024b), test-time fine-tuning methods (Kawar et al., 2023; Mokady et al., 2023), and training-free methods (Avrahami et al., 2022a; Parmar et al., 2023). Various conditions, such as text (Zhang et al., 2023b), masks (Xie et al., 2023), reference images (Zhang et al., 2023a), sketches (Cao et al., 2023), and layouts (Epstein et al., 2023), have been explored to guide the editing process.

Instruction-based image editing further extends this paradigm by allowing users to modify images using natural language instructions. These instructions may involve operations such as object insertion, removal, replacement, or attribute transformation. Compared with detailed text prompts, instructions provide a more natural and intuitive interface for image editing. A representative work is InstructPix2Pix (Brooks et al., 2023), which constructs large-scale instruction datasets through synthetic image-instruction pairs. Subsequent approaches attempt to improve editing quality by incorporating additional feedback signals (Zhang et al., 2024b) or simplifying instruction generation (Fu et al., 2023). Despite these advances, existing methods may struggle to accurately localize editing regions or correctly interpret complex instructions, especially when editing small objects or scenes with complicated spatial relationships.

2.2. Reinforcement learning for diffusion models

Reinforcement learning (RL) has recently emerged as an effective paradigm for aligning generative models with human preferences. In diffusion models, RL methods often formulate the denoising process as a sequential decision-making problem and optimize the generation process using reward signals. For example, DDPO (Black et al., 2023) models diffusion denoising as a multi-step Markov decision process and optimizes it with policy gradient methods. DPOK (Fan et al., 2024) further improves text-to-image diffusion models through online RL fine-tuning to enhance image-text alignment and visual quality. In addition, PRDP

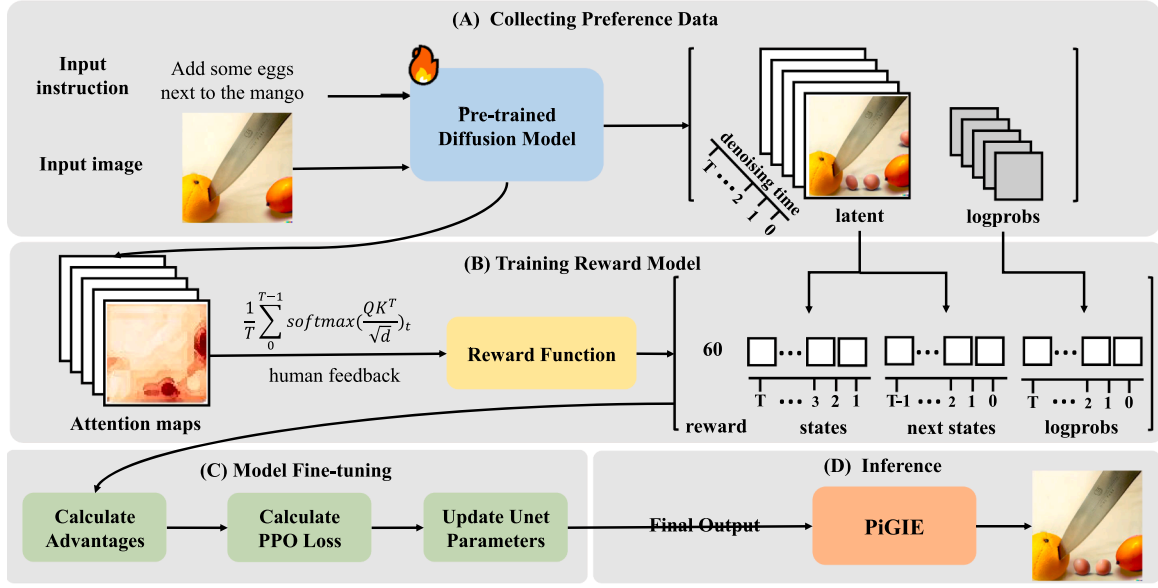


Fig. 2. Overview of PPO-Guided Image Editing (PiGIE), which uses Proximal Policy Optimization (PPO) to fine-tune the diffusion model-based image edit model. During training, in the first step, the input training images and instruction are used to generate latent, log probs, and attention maps using the pre-trained diffusion model. In the second step, the cosine similarity between the attention maps of UNet and the human-feedback true editing regions is computed as a reward function for the model. In the third step, the model is fine-tuned by PPO reinforcement learning using the reward model as a signal. During inferencing, our PiGIE accurately locates the edited region based on the instruction and achieves fine-grained local editing.

(Deng et al., 2024) introduces stable black-box reward optimization for diffusion models on large-scale datasets.

Proximal Policy Optimization (PPO) (Schulman et al., 2017) is a widely used policy gradient method that provides stable and efficient optimization in RL tasks. Recent studies have explored PPO-based optimization strategies to align generative models with human preferences or external reward signals.

While these works mainly focus on text-to-image generation, the application of reinforcement learning to instruction-based diffusion models for image editing remains relatively underexplored. In this work, we apply reinforcement learning to optimize cross-attention maps in the instruction-based image editing process using a PPO-based framework. By designing a reward signal based on the cosine similarity of attention maps and simulated human feedback, the proposed method improves the model's ability to localize editing regions and perform fine-grained edits in complex scenes.

3. Method

In this section, we introduce PiGIE, an RL fine-tuned instruction-based image editing diffusion model with significantly improved performance on complex image editing applications. As shown in Fig. 2, PiGIE takes both an original image and a human language instruction as input and outputs the predicted noise for the target image.

3.1. Reward function

We use a combination of two reward functions for the RL. The Attention Map Reward forces the diffusion model to learn correct editing regions, while the Clip reward assigns a constraint between the edited image and the input image.

Attention map reward. Given an input image $\mathcal{V}$, mask image $\mathcal{M}$ and an instruction $\mathcal{X}$, we want the editing model to generate an output image $\mathcal{O}$ that contains the target object specified by $\mathcal{X}$. We use the normalized mask image $\mathcal{M}$ from the MagicBrush dataset (Zhang et al., 2024a) to simulate human feedback and generate the attention map a_2 , which

shows the true position of the target object in $\mathcal{V}$. These masks simulate human-annotated editing regions. The masks are converted into attention maps and used as the target localization signal.

By computing the cross-attention between the encoded text and the intermediate features of the denoisers ϵ_θ , the pre-trained InstructPix2Pix model generates the reference attention map a_1 from $\mathcal{X}$.

$$\text{Attention}(Q, K, V) = M \cdot V \quad (1)$$

$$\text{where } M = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right) \quad (2)$$

Here, we are primarily concerned with the cross-attention map M , which is observed to be closely related to the structure of the image (Hertz et al., 2022). M_{ij} represents the weight of the value of the j th token on the pixel i . In addition, the cross-attention mask is specific to a certain time step, and we get a different attention mask M_t at each time step t . We define a_1 as the average of all tokens over the entire sampling time.

$$a_1 = \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{T} \sum_{t=0}^{T-1} M_t \right) \quad (3)$$

Where N is the number of tokens in the instruction $\mathcal{X}$, T is the diffusion steps.

The Attention Map Reward is defined as the cosine similarity between a_1 and a_2 , which measures how well the editing model can align the target object with the instruction.

$$L_{att} = \frac{a_1 \cdot a_2}{\|a_1\| \|a_2\|} \quad (4)$$

Such a reward can be viewed as a proxy of the perceptual quality of the output image, as it encourages the editing model to preserve the semantic content of the input image and modify only the relevant region.

Clip reward. Besides the Attention Map Reward, we also consider other reward that can improve the performance and stability of the RL-based image editing model.

Specifically, we use Clip Reward: a regularization term that penalizes the editing model for generating output images that are too different

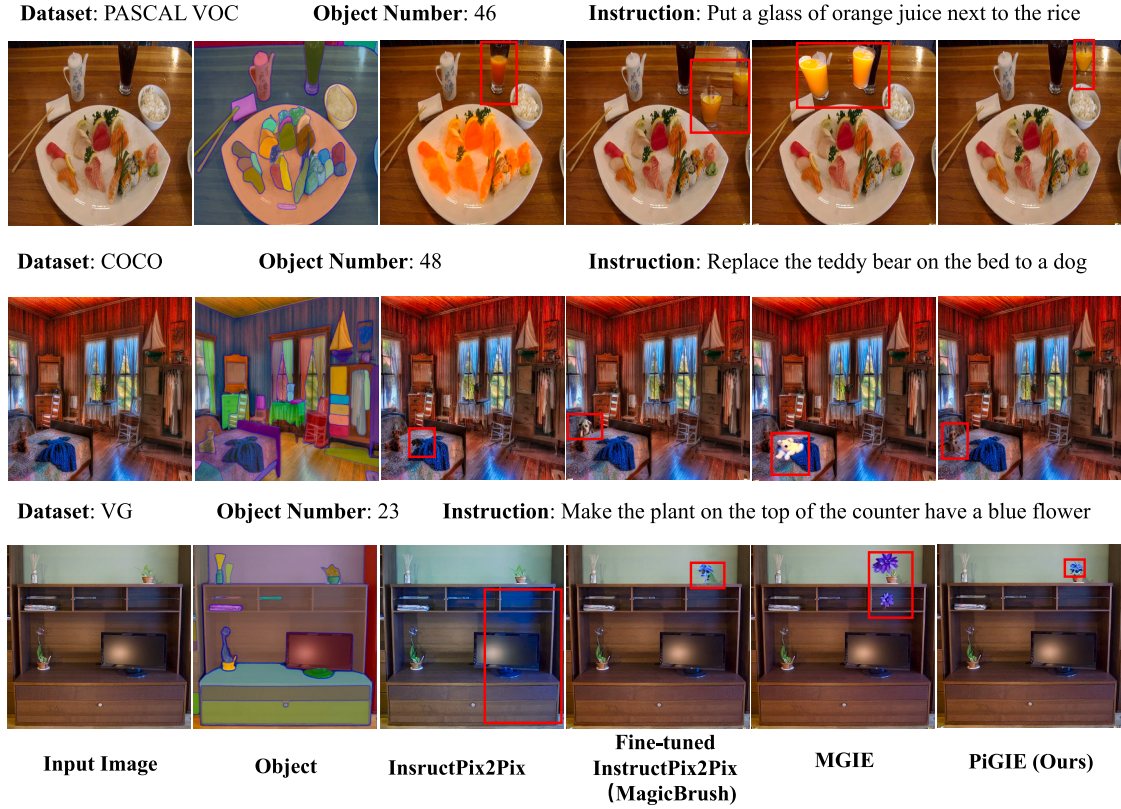


Fig. 3. Qualitative comparison between InstructPix2Pix, finetuned-InstructPix2Pix, MGIE and our PiGIE. Object number obtained by SAM (Kirillov et al., 2023) indicates that the image we select is complex enough. In the first example, the PiGIE accurately positioned the edit area to the edge of the rice. In the 2nd example, PiGIE pinpointed the location of the teddy bear, while the other three methods were unsuccessful. Furthermore, in the 3rd example, InstructPix2Pix was unable to localize the flower pot on the cabinet, while PiGIE accurately planted the blue flower on the flower pot.

from the input images in terms of pixel values. The Clip Reward is defined as the Mean Absolute Error (MAE) between the input image and the output image, clipped by a threshold τ :

$$L_{Clip} = \text{MAE}(\mathcal{V}, \mathcal{O}) \cdot \mathbb{I}(\text{MAE}(\mathcal{V}, \mathcal{O}) > \tau) \quad (5)$$

where $\mathbb{I}$ is an indicator function. The Clip Reward prevents the editing model from making drastic changes to the input image that may degrade the image quality or introduce artifacts.

Total reward function. The total reward function is a weighted sum of these rewards:

$$r(x_0, \mathcal{V}, \mathcal{X}) = L_{att} + \alpha \cdot L_{Clip} \quad (6)$$

Note that α is negative, as this term serves as a regularization penalty. By setting $\alpha < 0$, we discourage the model from deviating excessively from the input, balancing task performance with fidelity. We maximize this reward using the PPO algorithm as described in Section 3.2.

3.2. PPO-guided image editing (PiGIE)

Modeling editing model as Multi-Step MDPs. We get inspiration from DDPO (Black et al., 2023), which models the denoising process of the diffusion model as a multi-step Markov Decision Process (MDP) and optimizes the training of the diffusion model using the PPO (Schulman et al., 2017) algorithm. The Markov decision process can be formalized as $(S, \mathcal{A}, \rho_0, \mathcal{P}, \mathcal{R})$, where S is the state space, $\mathcal{A}$ is the action space, ρ_0 is the initial state distribution, $\mathcal{P}$ is the state transfer matrix, and $\mathcal{R}$ is the reward function. The optimization objective of reinforcement learning

is to maximize the cumulative payoff of the strategy:

$$J_{RL}(\pi) = \mathbb{E}_{\tau \sim p(\tau|\pi)} \sum_{t=0}^T \mathcal{R}(s_t, a_t) \quad (7)$$

We formulate the image editing task as an MDP, where the state is the input image $\mathcal{V}$, the action is a noise mask applied to the image, and the reward is the function mentioned in Section 3.1.

Following DDPO, we relate the diffusion modeling-based editing model multi-step denoising process to the MDP in the following way:

$$s_t \triangleq (\mathcal{V}, \mathcal{X}, t, x_t) \quad (8)$$

The state of each step is defined as a tuple containing the input image, the instruction, the denoising time step, and the denoising result of the current time step.

$$\pi(a_t | s_t) \triangleq p_{\theta}(x_{t-1} | x_t, \mathcal{V}, \mathcal{X}) \quad (9)$$

The strategy is the conditional distribution of the next denoising result given the current state.

$$A_t \triangleq x_{t-1} \quad (10)$$

The action is the result of the denoising in the next step.

$$P(s_{t+1} | s_t, a_t) \triangleq (\delta_{\mathcal{V}}, \delta_{\mathcal{X}}, \delta_{t-1}, \delta_{x_{t-1}}) \quad (11)$$

During the denoising process, the state transfer is deterministic after sampling the next denoising result x_{t-1} . Therefore, the state transfer probability is represented here by four Dirac delta distributions.

$$\rho_0(s_0) \triangleq (p(\mathcal{V}, \mathcal{X}), \delta_T, \mathcal{N}(0, \mathbb{I})) \quad (12)$$

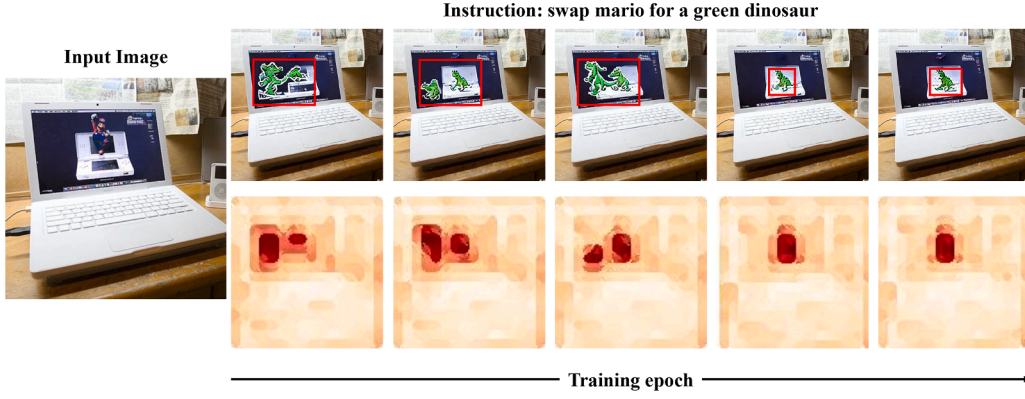


Fig. 4. Sample progressions of the same cue and random seeds during training. The attention map of the samples will tend to localize more faithfully to the correct editing region.

Table 1

Quantitative comparison of image editing baselines on the MagicBrush test set. Evaluation metrics include pixel-wise L1 and L2 errors (lower is better). The best results are highlighted in bold.

	Methods	L1↓	L2↓	CLIP-I↑	DINO↑
Global Description-guided	Open-Edit (Liu et al., 2020)	0.1430	0.0431	0.8381	0.7632
	SD-SDEdit (Meng et al., 2021)	0.1014	0.0278	0.8526	0.7726
	Null Text Inversion (Mokady et al., 2023)	0.0749	0.0197	0.8827	0.8206
Mask-guided	Blended Diffusion (Avrahami et al., 2022b)	3.5631	119.2813	0.9291	0.8644
Instruction-guided	InstructPix2Pix (Brooks et al., 2023)	0.1122	0.0371	0.8524	0.7428
	Fine-tuned InstructPix2Pix(MagicBrush) (Zhang et al., 2024a)	0.0625	0.0353	0.8924	0.8273
	MGIE (Fu et al., 2023)	0.07868	0.02984	0.9141	0.8536
	ICEdit (Zhang et al., 2026)	0.0600	–	0.9280	0.8530
	EDITMGT (Chow et al., 2026)	0.0580	–	0.9110	0.8810
	PiGIE(Ours)	0.0576	0.0172	0.9400	0.9046

For the initial state distribution, the condition variable obeys its prior distribution, the time step T is determined, and finally x_t obeys standard Gaussian noise.

$$R(s_t, a_t) \triangleq \begin{cases} r(x_0, \mathcal{V}, \mathcal{X}) & t = 0 \\ 0 & \text{otherwise} \end{cases} \quad (13)$$

During the denoising process, only the final denoising result receives a score based on the reward model mentioned above, while the reward values during the denoising process are all defined as 0.

Finally, we align the denoising process of the diffusion model-based editing model with that of the MDP so that the optimization objective $\mathcal{J}_{RL}(\pi)$ Equivalent to $\mathcal{J}_{PiGIE}(\pi)$.

$$\mathcal{J}_{PiGIE}(\pi) = \mathbb{E}_{\mathcal{V} \sim p(\mathcal{V}), \mathcal{X} \sim p(\mathcal{X}), x_0 \sim p_{\theta}(x_0 | (\mathcal{V}, \mathcal{X}))} [r(x_0, \mathcal{V}, \mathcal{X})] \quad (14)$$

Policy Gradient Estimation. To optimize, it is then necessary to estimate its gradient. We use PPO for gradient estimation of $\mathcal{J}_{PiGIE}(\pi)$:

$$\nabla \mathcal{J}_{PiGIE}(\pi) = \mathbb{E} \left[\left(\text{CLIP} \left(\frac{p_{\theta}(x_{t-1} | x_t, \mathcal{V}, \mathcal{X})}{p_{\theta_{old}}(x_{t-1} | x_t, \mathcal{V}, \mathcal{X})}, 1 - \epsilon, 1 + \epsilon \right) \cdot \nabla \log p_{\theta}(x_{t-1} | x_t, \mathcal{V}, \mathcal{X}) \cdot r(x_0, \mathcal{V}, \mathcal{X}) \right) \right] \quad (15)$$

Once the estimates are obtained, an optimization algorithm can be applied to optimize the parameters of the diffusion model to maximize.

4. Experiment

In this section, we conduct a series of comprehensive experiments to show the superior performance of PiGIE, with a special emphasis on editing tiny objects.

Table 2

Ablation study comparing different hyperparameters.

α	τ	L1↓	L2↓	CLIP-T↑	SSIM↑	PSNR↑
-0.5	100	0.0576	0.0172	0.2353	0.8146	25.4742
-1	100	0.0608	0.0189	0.2391	0.7978	24.2575
-2	100	0.0603	0.0190	0.2370	0.7973	21.0143
-1	50	0.0616	0.0197	0.2370	0.7911	23.4662
-1	200	0.0652	0.0207	0.2384	0.7753	22.9811

Table 3

Ablation study comparing different reward functions.

Methods	L1↓	L2↓	CLIP-T↑	SSIM↑	PSNR↑
Attn-Only	0.0601	0.0184	0.2380	0.7545	20.0518
Clip-Only	0.0588	0.0167	0.2362	0.7596	20.3118
PiGIE	0.0576	0.0172	0.2353	0.8146	25.4742

4.1. Implementation details

Datasets and evaluation metrics. We use MagicBrush (Zhang et al., 2024a) as our pre-training data. It comprises 10K (source image, instruction, target image) triples, covering diverse scenarios single-turn, multi-turn, mask-provided, and mask-free editing. We use the source image and instructions from the dataset. Then, to simulate human feedback, we choose a free-form mask image from the MagicBrush dataset to simulate human feedback on the editing area of the image.

For a comprehensive evaluation, we consider various editing aspects. We selected 3 global description-guided image editing models: Open-Edit (Liu et al., 2020), SD-SDEdit (Meng et al., 2021), and Null Text Inversion (Mokady et al., 2023), 1 mask-guided image editing methods:

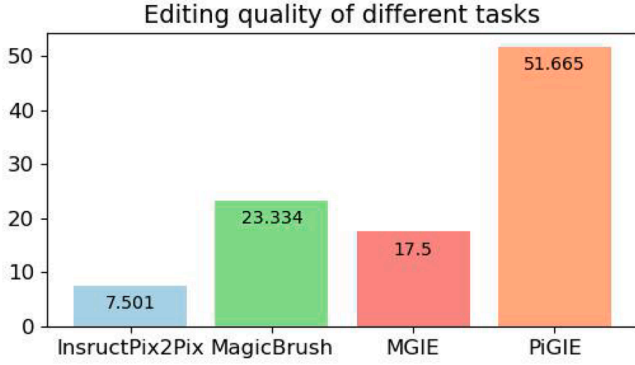


Fig. 5. Human evaluation: editing quality of different tasks.

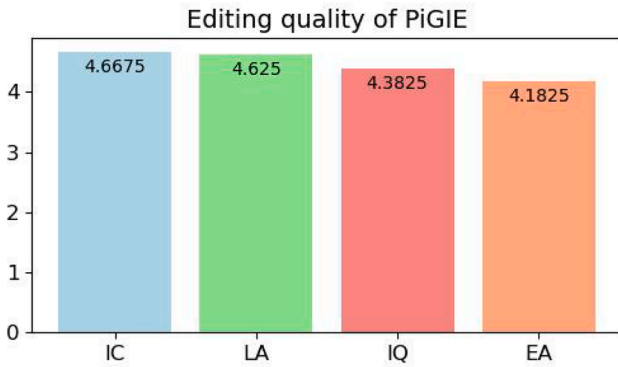


Fig. 6. Human evaluation: editing quality of PiGIE.

Blended Diffusion (Avrahami et al., 2022b) and 3 existing instruction-based image editing models: InstructPix2Pix (Brooks et al., 2023), Fine-tuned InstructPix2Pix (Zhang et al., 2024a) and MGIE (Fu et al., 2023) to compare with our results. All methods are evaluated on MagicBrush test set. We use the L1 and L2 to measure the standard pixel difference between the generated image and the ground truth image. Additionally, we use CLIP-I (Radford et al., 2021) and DINO (Caron et al., 2021) to measure the cosine similarity between the generated image and ground truth image.

As for test datasets, we choose 3 image datasets with complex scenarios: PASCAL VOC (Everingham et al., 2010), COCO (Lin et al., 2014) and VG (Krishna et al., 2017). After slicing and dicing the image using Segment Anything (Kirillov et al., 2023), we randomly selected images with more than 20 objects for the qualitative study because the more objects in the diagram, the more complex the image and the more difficult the positional relationships. Therefore, we choose the images in these three datasets as our input images, and due to the lack of editing instructions, we use Shikra (Chen et al., 2023) to assist us in understanding the image information and generating appropriate editing instructions. All evaluations are in a zero-shot manner, where the model is only pre-trained on MagicBrush. We compare the edited image with four instruction-based methods: InstructPix2Pix (Brooks et al., 2023), Fine-tuned InstructPix2Pix (Zhang et al., 2024a), MGIE (Fu et al., 2023) and our PiGIE.

Baselines. We treat several instruction-based image editing models, such as InstructPix2Pix (Brooks et al., 2023), as our baseline. We also consider the MLLM-guided image editing (MGIE) model, which uses LLaVA-7B (Liu et al., 2024) and StableDiffusion-v1.5 (Rombach et al., 2022) to edit the image.

4.2. Evaluations

We next evaluate the ability of PiGIE to fine-tune UNet. We provide the relative effectiveness of the reward function during training in Appendix B.1. The gradual increase in the reward function illustrates that our model has the ability to adjust the attention map to localize regions through training.

The progression of finetuning is depicted in Fig. 4. Qualitatively, during training, the attention map of the samples will tend to localize more faithfully to the correct editing region. After fine-tuning, our model can generate editing effects that match human cognition based on human commands.

4.3. Comparison

Compare with other tasks. We first evaluate all methods on existing instruction based image editing tasks: InstructPix2Pix, fine-tuned InstructPix2Pix, MGIE and our PiGIE. All methods are evaluated on the MagicBrush test set. Table 1 shows the results of the instruction-based image editing methods. We have the following observations: (1) In editing scenarios with complex datasets, existing InstructPix2Pix and other instruction-based image editing methods are more competitive with global description-guided methods such as Open-Edit (Liu et al., 2020), SD-SDEdit (Meng et al., 2021), and Null Text Inversion (Mokady et al., 2023). (2) After fine-tuning with the MagicBrush dataset (Zhang et al., 2024a), InstructPix2Pix shows some improvement in performance on most of the metrics. (3) Since our PiGIE uses mask image as an input condition, for a fair comparison, we compare PiGIE with Blended Diffusion (Avrahami et al., 2022b), which is one of the most recent mask-conditioned image editing models. As we can see, our PiGIE model performs well on global editing metrics (CLIP-I, DINO) and excellent on local editing metrics (L1, L2). Therefore we conclude that PiGIE is a strong yet highly robust model across multiple metrics. (4) In Appendix B.2, we also show more editing results of our PiGIE on the style editing task and the fine editing task, respectively. Our PiGIE has a lower L1 and L2 and higher CLIP-I and DINO compared with the existing methods, which is because we use Clip Reward to limit the reward function, which makes the edited image retain most of the features of the original image.

Compare with different datasets. Fig. 3 illustrates a visual comparison of all the datasets used. Object number obtained by SAM (Kirillov et al., 2023) indicates that the image we select is complex enough. We selected several images from different datasets for the zero-shot test, the results show that our PiGIE edits are significantly better than both InstructPix2Pix and MGIE on different complex image datasets. Our PiGIE is more accurate in understanding the positional relationship (example1) and the quality of editing is higher (example2, example3).

4.4. Ablation study

PPO-Guided Image Editing has shown encouraging improvements in the precise editing of complex scenes. Now, we intend to evaluate the different architectures of the model as well as some important hyperparameters.

Reward function. Table 3 considers Attn-Only, Clip-Only, and our PiGIE. Attn-Only speaks only of Attention Map Reward as the reward function of the model. In contrast, Clip-Only uses only Clip Reward as a reward function. The experimental results show that Clip-Only has the lowest L2, which is because using Clip Reward as a reward function causes the model to more easily generate images that are similar to the original image and lose the editing function. Attn-Only has the highest CLIP-T, which might be because training using only Attention Map Reward increases the correlation between instruction and image. SSIM and PSNR metrics with Attention Map Reward only are not as good as those of our

PiGIE, which indicates that the linear combination of Attention Map Reward and Clip Reward enables the model to successfully achieve fine editing of complex images while preserving the features of the original image.

Hyperparameters. There are two important hyperparameters in the training process of PiGIE: α controls the weight of Clip Reward in the reward function, and τ controls the threshold for clipping Clip Reward. Table 2 shows the evaluation metrics of the model on L1, L2, CLIP-T, SSIM, and PSNR at different α and τ , Appendix B.3 shows the relative effectiveness of the reward function with different hyperparameters. The result shows that all indicators are optimal when $\alpha = -0.5$ and $\tau = 100$. The possible reasons for this phenomenon are (1) the model is more sensitive to the changes in the Attention Map Reward during training and thus needs a lower α value to allow the model to optimize the Attention Map Reward at a deeper level, thus improving the alignment of the attention map. (2) The model needs both Clip Reward to prevent the editing model from changing the original map too much and not to lose the editing ability, so τ choosing an intermediate value will better improve the generalization of the model. Therefore, we finally choose $\alpha = -0.5$, $\tau = 100$ as the training parameter for our model.

Human evaluation. We conducted a human evaluation to assess the image editing results. First, 8 randomly selected images from the dataset were edited using instructPix2Pix, MGIE, and our PiGIE, with rankings determined by human evaluators. As shown in Fig. 5, 51.665% of participants rated PiGIE as having the best editing capabilities, significantly outperforming MagicBrush (23.334%), which ranked second. This suggests that PiGIE is widely considered superior to other command-based image editing models in terms of editing ability.

Next, 4 images were randomly selected for inference using PiGIE, and human evaluators assessed the model on 4 metrics: instruction comprehension, localization of the editing region, image quality, and overall editing ability. Fig. 6 illustrates the results. The highest scores were for instruction comprehension (IC) and localization accuracy (LA) (4.6675/5 and 4.625/5, respectively), while image quality (IQ) and editing ability (EA) also received strong scores (4.3875/5 and 4.1875/5). These results indicate general satisfaction with PiGIE, which effectively preserves original image features and accurately localizes editing regions. Detailed results are in Appendix B.4.

Inference efficiency. Our PiGIE are TEXT-ONLY during inference, using only natural human instructions to edit the images. As a result, our speed in the inference process has a great advantage over the MGIE model using MLLM(9.2s). Compared to InstructPix2Pix(6.8s), our model also takes only 8.9 s to complete the inference on NVIDIA 3090 GPUs.

Qualitative comparisons. Appendix B.2 shows the editing results of our PiGIE on the style editing task and the fine-grained editing task, respectively.

5. Conclusion

We present PPO-Guided Image Editing (PiGIE), an instruction-based model designed to precisely identify editing regions, especially in complex scenes with multiple objects and intricate positional relationships. The success of PiGIE not only establishes a new benchmark for instruction-based image editing but also opens up a completely new path for the application of reinforcement learning in image editing.

CRedit authorship contribution statement

Tiancheng Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Jinxiu Liu:** Writing – review & editing, Methodology, Investigation, Formal analysis; **Weijian Luo:** Writing – review & editing; **Huajun Chen:** Investigation, Data curation; **Qi Liu:** Writing – review & editing, Methodology, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Qi Liu reports financial support was provided by GJYC program of Guangzhou under Grant 2024D01J0081. Qi Liu reports financial support was provided by ZJ program of Guangdong under Grant 2023QN10X455. Qi Liu reports financial support was provided by Fundamental Research Funds for the Central Universities under Grant 2025ZYGXZR053. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Model Architecture

A.1. Background: proximal policy optimization (PPO)

PPO (Schulman et al., 2017) is an actor-critic RL algorithm that alternates between sampling trajectories from the current policy and updating the policy using a clipped surrogate objective. Let π_θ be a policy, the PPO trains the policy by minimizing the objective:

$$L_{PPO}(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^T \min \left(\rho_t(\theta) A_t, \text{Clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t \right) \right] \quad (\text{A.1})$$

where θ are the policy parameters, $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$ is the probability ratio, A_t is an estimate of the advantage function, and ϵ is a clipping parameter. The operation Clip represents clipping the values.

The advantage function measures how much better an action is compared to the average action at a given state, and can be computed using a value network that estimates the state value function $V(s_t)$. The value network shares the same architecture as the policy network, except that it outputs a scalar value instead of a noise mask.

A.2. Training details

We treat DDPO (Black et al., 2023) as our base code. The instruction-based diffusion model $\mathcal{F}$ is initialized from InstructPix2Pix (Brooks et al., 2023). We train the UNet with low-rank adaptation (LoRA) (Shen et al., 2021). Following InstructPix2Pix (Brooks et al., 2023), we concatenate the input image with added noise, which makes our $\mathcal{F}$ condition both $\mathcal{V}$ and $\mathcal{X}$ during image editing. We also take classifier-free guidance (Ho & Salimans, 2022) to trade off the quality and diversity of samples generated by a diffusion model. We adopt AdamW (Loshchilov & Hutter, 2017) with a batch size of 8 to optimize PiGIE. The learning rates of the $\mathcal{F}$ is $2e-4$, training epoch is 200, and the hyperparameter α is -0.5, τ is 100, respectively. All experiments are conducted in PyTorch on 4 3090 GPUs.

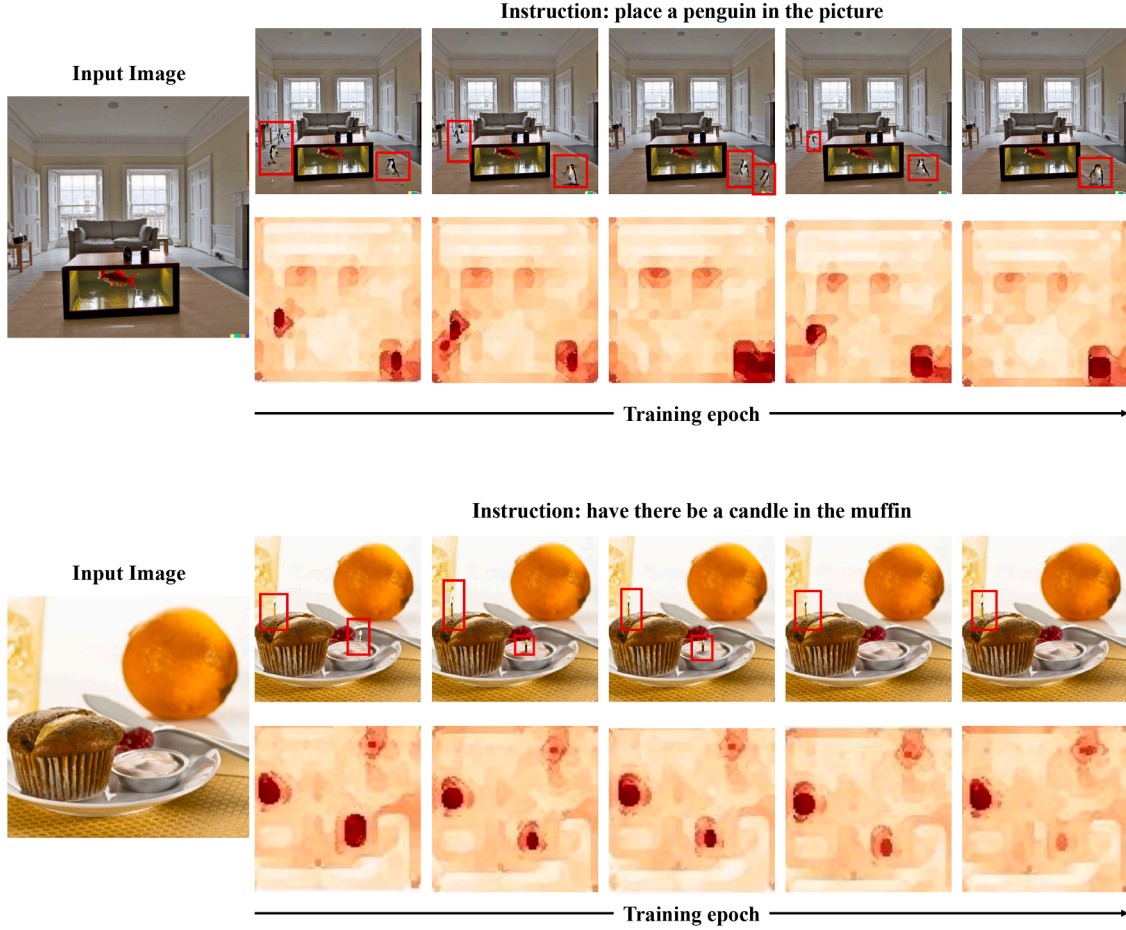


Fig. A.7. More sample progressions of the same cue and random seeds during training. The attention map of the samples will tend to localize more faithfully to the correct editing region.

Table A.4
Ablation study of human evaluation.

	Methods	InstructPix2Pix	Fine-tuned InstructPix2Pix(MagicBrush)	MGIE	PiGIE(Ours)
Editing quality of different tasks	image1	6.67	16.67	13.33	63.33
	image2	3.33	20	13.33	63.33
	image3	6.67	43.33	16.67	33.33
	image4	26.67	16.67	36.67	20
	image5	3.33	6.67	20	70
	image6	6.67	36.67	3.33	53.33
	image7	6.67	13.33	26.67	53.33
	image8	0	33.33	10	56.67
	mean value	7.501	23.334	17.500	51.665
Editing quality of PiGIE	Metrics	IC	LA	IQ	EA
	image1	4.27	4.5	4.33	4.1
	image2	4.63	4.63	4.37	4.33
	image3	4.9	4.57	4.6	4.3
	image4	4.87	4.8	4.23	4.18
	mean value	4.6675	4.625	4.3825	4.1825

A.3. Evaluation details

We totally use 7 metrics to evaluate the model: L1, L2, CLIP-I, DINO, CLIP-T, SSIM, and PSNR.

L1. L1 measures the absolute pixel-wise differences between the predicted values and the ground truth. It's calculated as the mean of the absolute differences between corresponding pixels in the predicted and

ground truth images.

$$L1 = \frac{1}{N} \sum_{i=1}^N |p_i - g_i|$$

Here, p_i and g_i are the pixel values of the predicted and ground truth images respectively, and N is the total number of pixels.

L2. L2 measures the squared differences between the predicted values and the ground truth. It's calculated as the mean of the squared differ-

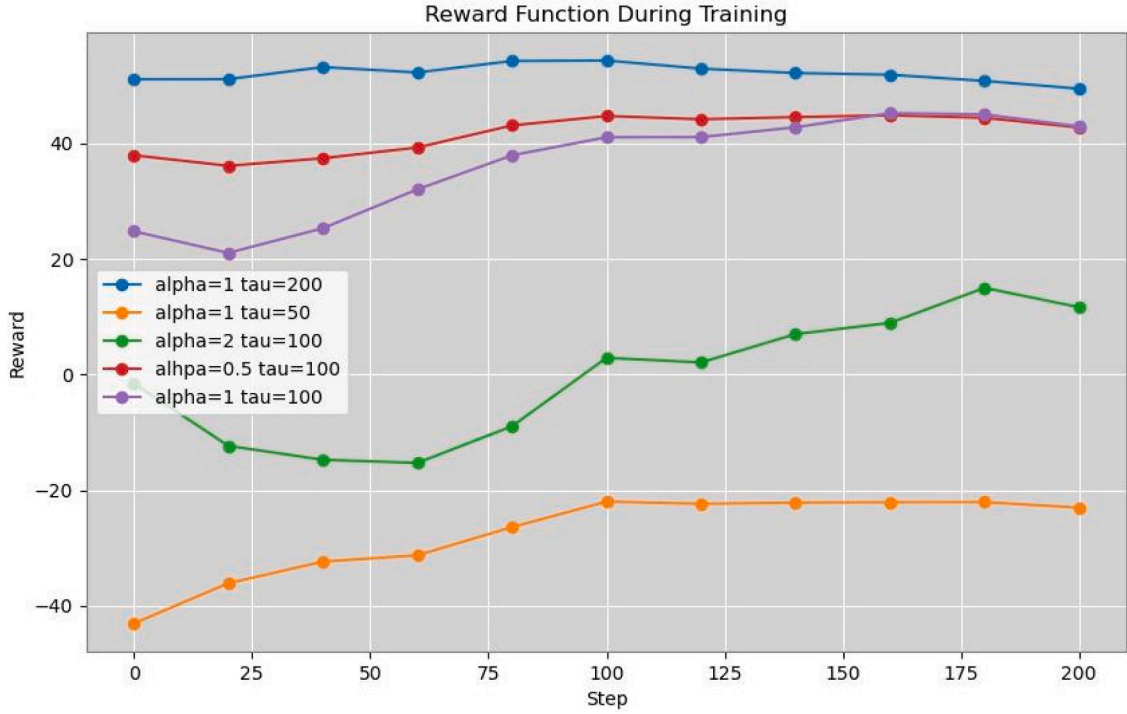


Fig. A.8. Plot of the reward function as a function of the training process when the hyperparameters α , τ take different values.

ences between corresponding pixels in the predicted and ground truth images.

$$L2 = \frac{1}{N} \sum_{i=1}^N (p_i - g_i)^2$$

CLIP-I. CLIP-I is the average pairwise cosine similarity between CLIP embeddings of generated and real images.

$$\text{CLIP-I} = \frac{1}{N} \sum_{i=1}^N \cos_sim(\text{CLIP}(G_i), \text{CLIP}(R_i)) \quad (\text{A.2})$$

Where:

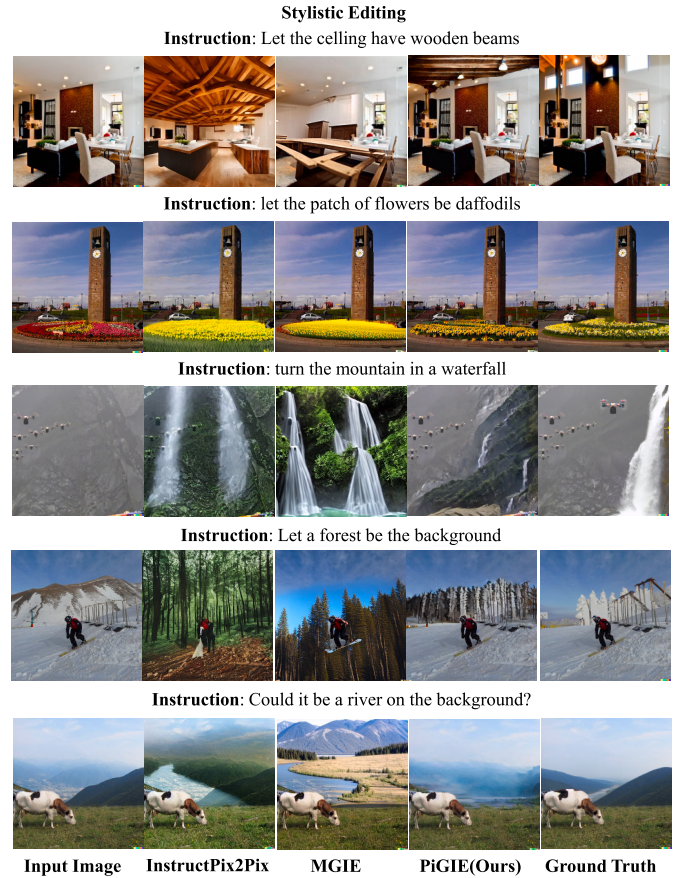
- N is the total number of image pairs.
- $\cos_sim(x, y)$ denotes the cosine similarity function between vectors x and y .
- $\text{CLIP}(G_i)$ represents the CLIP embedding of the generated image i .
- $\text{CLIP}(R_i)$ represents the CLIP embedding of the corresponding real image i .

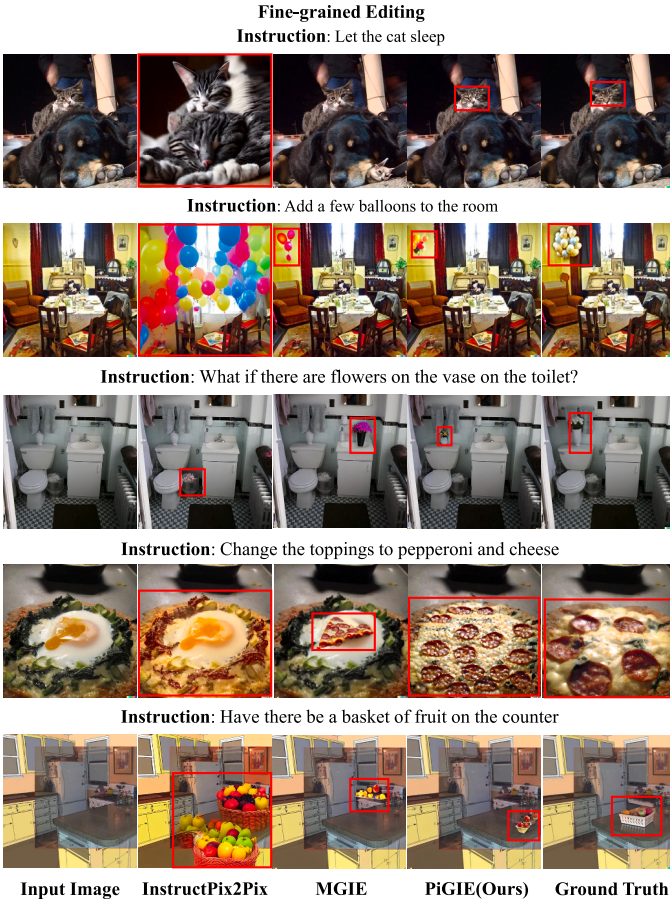
CLIP-T. The second important aspect to evaluate is prompt fidelity, measured as the average cosine similarity between prompt and image CLIP embeddings.

$$\text{CLIP-T} = \frac{1}{N} \sum_{i=1}^N \cos_sim(\text{CLIP}(P), \text{CLIP}(I_i)) \quad (\text{A.3})$$

Where:

- Prompt Fidelity represents the prompt fidelity, measured as the average cosine similarity between prompt and image CLIP embeddings.
- N is the total number of images.
- $\cos_sim(x, y)$ denotes the cosine similarity function between vectors x and y .
- $\text{CLIP}(P)$ represents the CLIP embedding of the prompt.
- $\text{CLIP}(I_i)$ represents the CLIP embedding of the i^{th} image.





SSIM. SSIM (Structural Similarity Index) is a metric used to measure the similarity between two images, considering aspects like luminance, contrast, and structure. It evaluates how perceptually similar the images are, taking into account human visual perception. By comparing the brightness, contrast, and structural information at different scales, SSIM provides a comprehensive similarity measure ranging from 0 to 1, where a value closer to 1 indicates higher similarity.

PSNR. PSNR measures the quality of a reconstructed signal. It compares the maximum possible power of a signal with the power of corrupting noise.

$$PSNR = 10 \cdot \log_{10} \left(\frac{\max \text{ pixel value}^2}{MSE} \right)$$

Here, MSE is the Mean Squared Error calculated earlier.

A.4. Human evaluation details

Our human evaluation questionnaire can be accessed by clicking on the following link: <https://www.wenjuan.com/s/UZBZJv1x7p/>

Appendix B. Additional results

B.1. More results for the sample progressions during training

More progression of finetuning is depicted in Fig. A.7. Qualitatively, during training, the attention map of the samples will tend to localize more faithfully to the correct editing region. For example, we use the edit instruction "place a penguin in the picture", during training, the attention map of the samples goes from being initially localized to the left and right to being specified to be located on the right, and the editing effect goes from increasing the penguins on both the left and right to increasing them only on the right.

B.2. More results for evaluations

We show the editing results of our PiGiE on the style editing task and the fine editing task, respectively. Compared to InstructPix2Pix and MGIE, our model not only does not lose to existing models in style editing but also shows strong editing region localization and local editing ability in fine editing.

B.3. More results for hyperparameters ablation study

Fig. A.8 shows the reward function as a function of the training process when the hyperparameters α , τ take different values. From this figure, it can be seen that all the reward functions show an upward trend, but when τ is too large, the upward trend becomes very slow, and when α takes a smaller value, the reward function has a higher average value. Therefore, we combine the altered graph with the metrics and choose $\alpha = -0.5$ and $\tau = 100$ as our hyperparameters.

B.4. More results for human evaluations

We conducted a user study to evaluate both the overall editing preference across methods and the editing quality of PiGiE. To avoid bias, all edited results were randomly shuffled before evaluation.

For the editing quality of different tasks, we selected 8 representative examples and compared the outputs generated by InstructPix2Pix, Fine-tuned InstructPix2Pix (MagicBrush), MGIE, and PiGiE. Participants were provided with the source image, editing instruction, and edited results from all methods, and were asked to select the result that best satisfied the editing instruction while preserving overall visual quality.

The percentages reported in Table A.4 indicate the proportion of votes received by each method.

For the **editing quality of PiGiE**, we further selected 4 representative examples and asked participants to score PiGiE using a 5-point Likert scale (1 = poor, 5 = excellent) according to the following criteria:

- **IC (Instruction Consistency):** whether the edited result correctly follows the given editing instruction.
- **LA (Localization Accuracy):** whether modifications are restricted to the intended region without affecting unrelated regions.
- **IQ (Image Quality):** the realism and overall visual quality of the edited image.
- **EA (Editing Accuracy):** whether the target attributes are edited accurately and completely.

The final score is calculated as the average across all participants. As shown in Table A.4, PiGiE achieves consistently high scores across all dimensions, especially in instruction consistency and localization accuracy.

Appendix C. Limitation and future work

In this paper, we use PPO reinforcement learning to fine-tune the diffusion model to achieve fine-grained local editing. Although our work facilitates creative visual applications, these limitations should still be considered when interpreting the results. Firstly, Due to the limitations of the text encoder, our model cannot understand positional relations such as left, right, etc. For example, we cannot use the command "turn the dog on the left into a cat" to make the model localize only the dog on the left. Secondly, the model lacks the ability of language base and numerical perception. We will investigate these directions in the future to realize more practical and powerful instruction-based image editing.

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